# ML & DL ALGORITHMS

## PART I: CLASSICAL PARADIGM & LEARNING TAXONOMY

**Technical Reference Compendium**
Scope: Classical ML, DL & Metrics
Format: 3-Page Reference Guide

**Executive Overview:** Machine Learning (ML) algorithms construct mathematical models based on sample training data to make predictions or decisions without explicit rule programming. The paradigm spans Supervised Learning (mapping inputs to known targets), Unsupervised Learning (discovering latent structures), and Reinforcement Learning (optimizing sequential action policies). Selecting appropriate algorithms and matching evaluation metrics is fundamental to robust model deployment.

## 1. MACHINE LEARNING PARADIGM ARCHITECTURE

### Supervised
Labeled Data

Learns an explicit mapping function $f: X \rightarrow Y$. Used for Classification (discrete targets) and Regression (continuous numeric targets). Requires gold-standard ground truth annotations.

### Unsupervised
Unlabeled Data

Discovers hidden patterns, clusters, or lower-dimensional representations in unannotated data $X$. Key areas include Clustering, Density Estimation, and Dimensionality Reduction.

### Reinforcement
Reward-Based

Agents interact with an environment to maximize cumulative scalar reward $R_t$ via trial-and-error policy iteration governed by Markov Decision Processes (MDPs).

## 2. CORE CLASSICAL ML ALGORITHMS

| Algorithm | Category | Core Mechanism & Objective | Primary Strengths |
|---|---|---|---|
| **Linear / Logistic Regression** | Supervised | Fits linear combinations $w^T x + b$; uses Log-Loss for binary classification probabilities. | High interpretability, baseline speed, low computational overhead. |
| **Support Vector Machines (SVM)** | Supervised | Finds optimal maximum-margin hyperplane; projects non-linear data via kernel trick. | Effective in high-dimensional spaces; robust against overfitting. |
| **Decision Trees / Random Forests** | Supervised | Splits feature space using Information Gain / Gini Impurity; ensembles decrease variance. | Handles non-linearities, missing data, and mixed feature types. |
| **Gradient Boosting (XGBoost/LightGBM)** | Supervised | Sequentially fits weak decision tree learners to the negative gradient of the loss function. | State-of-the-art accuracy on tabular data; handling of sparse features. |
| **K-Means Clustering** | Unsupervised | Iteratively shifts $k$ centroids to minimize intra-cluster Within-Cluster Sum of Squares (WCSS). | Computationally efficient, scales linearly to large datasets $O(n)$. |
| **Principal Component Analysis (PCA)** | Unsupervised | Orthogonal linear transformation projecting data onto axes of maximum variance. | Reduces dimensionality, removes collinearity, aids visual exploration. |

## 3. ALGORITHM EXECUTION WORKFLOW



# ML & DL ALGORITHMS

## PART II: DEEP LEARNING ARCHITECTURES & NEURAL NETS

**Technical Reference Compendium**
Scope: Deep Neural Networks
Format: 3-Page Reference Guide

Deep Learning (DL) leverages hierarchical neural network layers to learn compositional feature representations directly from raw data. By substituting hand-engineered features with end-to-end backpropagation optimization, DL achieves state-of-the-art capability in vision, sequence modelling, language understanding, and generative synthesis.

## 1. NEURAL NETWORK ARCHITECTURAL SPECTRUM

### Convolutional Networks (CNNs)

**Computer Vision**

Employs spatial parameter sharing through kernel convolution filters, pooling operations, and translation equivariance. Standard backbone for image classification, object detection (YOLO), and semantic segmentation.

### Recurrent Networks (RNN / LSTM)

**Sequential**

Processes sequential inputs by maintaining recurrent hidden states $h_t$. Long Short-Term Memory (LSTM) units incorporate input, forget, and output gates to mitigate vanishing gradients over extended time steps.

### Transformer Architectures

**Attention-Based**

Eliminates recurrence entirely using Scaled Dot-Product Multi-Head Self-Attention. Enables parallelized training over vast sequences. Underpins Large Language Models (LLMs) and Vision Transformers (ViT).

### Generative Diffusion Models

**Generative AI**

Learns generative distributions by modeling a forward Gaussian noise addition process and optimizing a neural network (U-Net) to execute step-wise reverse denoising trajectories (e.g., Stable Diffusion).

## 2. MATHEMATICAL FORMULATIONS OF KEY MECHANISMS

**Scaled Dot-Product Attention:**
Given Query ($Q$), Key ($K$), and Value ($V$) matrices with feature dimension $d_k$:

$$Attention(Q, K, V) = softmax( (Q K^T) / \sqrt{d_k} ) V$$

**LSTM Gate Architecture:**

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad | \quad i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad | \quad C_t = f_t * C_{t-1} + i_t * tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

## 3. DEEP LEARNING ARCHITECTURE COMPARISON

| Architecture Type | Key Structural Innovations | Primary Loss Functions | Dominant Application Fields |
|---|---|---|---|
| **Deep Feedforward (MLP)** | Fully connected dense layers, non-linear activations (ReLU, GELU). | MSE, Cross-Entropy Loss | Tabular processing, simple classification. |
| **Convolutional (ResNet, ConvNeXt)** | Residual skip connections $y = F(x) + x$ preventing degradation. | Cross-Entropy, Focal Loss | Medical imaging, autonomous driving vision. |
| **Transformers (Decoder-Only)** | Causal masked attention, Rotary Position Embeddings (RoPE). | Autoregressive Log-Likelihood | Generative text (GPT-4, Llama 3), code synthesis. |
| **Autoencoders / VAEs** | Latent bottleneck layer; ELBO variational optimization. | Reconstruction Loss + KL-Divergence | Dimensionality reduction, anomaly detection. |

# ML & DL METRICS

## PART III: COMPREHENSIVE MODEL EVALUATION METRICS

**Technical Reference Compendium**
Scope: Quantitative Metrics
Format: 3-Page Reference Guide

Evaluation metrics provide quantitative measures of a model's operational performance. Selecting metrics strictly aligned with problem domain properties (such as class imbalance, extreme outliers, or ranking position preference) is critical to prevent misleading performance assessments.

## 1. CLASSIFICATION & REGRESSION METRICS

| Metric Name | Mathematical Formulation | Primary Use Case & Nuance |
|---|---|---|
| **Accuracy** | $(TP + TN) / (TP + TN + FP + FN)$ | Balanced datasets only; highly misleading under severe class imbalance. |
| **Precision & Recall** | $Prec = TP / (TP + FP)$ \| $Rec = TP / (TP + FN)$ | Precision minimizes False Positives; Recall minimizes False Negatives (e.g., medical). |
| **F1-Score** | $2 \cdot (Precision \cdot Recall) / (Precision + Recall)$ | Harmonic mean balancing precision and recall on imbalanced datasets. |
| **ROC-AUC** | Area under True Positive Rate vs. False Positive Rate curve | Evaluates ranking/discrimination capacity across all classification thresholds. |
| **Mean Squared Error (MSE)** | $MSE = (1/n) \sum (y_i - \hat{y}_i)^2$ | Regression metric heavily penalizing large error outliers due to squaring term. |
| **Mean Absolute Error (MAE)** | $MAE = (1/n) \sum \|y_i - \hat{y}_i\|$ | Robust regression metric providing linear penalty scale for intuitive error unit interpretation. |

## 2. CLUSTERING, RANKING & GENERATIVE METRICS

### Silhouette Coefficient — Clustering

Measures how well-separated clusters are: $s = (b - a) / max(a, b)$, where $a$ is intra-cluster distance and $b$ is nearest-cluster distance. Ranges from -1 to +1.

### NDCG@K — Ranking

Normalized Discounted Cumulative Gain evaluates search/recommendation relevance by discounting items appearing lower in the ranked list by logarithmic factor $log_2(i+1)$.

### BLEU / ROUGE — NLP Generation

BLEU measures precision of n-gram overlaps against reference translations. ROUGE measures recall of n-grams, serving as standard for summarization systems.

### Fréchet Inception Distance (FID) — Image Generation

Calculates Wasserstein-2 distance between feature embeddings of real vs. generated images extracted from an Inception-v3 network. Lower scores indicate superior realism.

## 3. METRIC SELECTION DECISION MATRIX

**Deployment Metric Selection Rules:**

- **Imbalanced Classification:** Prioritize Precision-Recall AUC (PR-AUC) or F1-score over Standard Accuracy or ROC-AUC.
- **Regression with Heavy Outliers:** Use Huber Loss or MAE rather than MSE or Root Mean Squared Error (RMSE).
- **Generative LLMs:** Pair automated metrics (BLEU/ROUGE) with LLM-as-a-Judge pairwise Elo scoring and human evaluation rubrics.